

Nicholas Le

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EDUCATION

University of California, San Diego
B.S. in Computer Engineering

Aug. 2024 – June 2028
San Diego, CA

WORK EXPERIENCE

Undergraduate Research Assistant

Apr. 2026 – Present

System Energy Efficiency (SEE) Lab, UC San Diego (PI: Prof. Tajana Rosing)

San Diego, CA

- Reproduced the published HyperSpec GPU spectral-clustering benchmark (PXD000561, ~7.46M clusters) to within 0.001% of the reference run, after porting its **CUDA kernels** from Numba to **CuPy RawKernels** to enable execution on **NVIDIA Blackwell (sm_120)** GPUs.
- Extended **HyperOMS**, a GPU open-modification search engine built on **Hyperdimensional Computing (HDC)**, to metabolomics — running spectral library search against ~1.5M positive-mode GNPS spectra and resolving a **ZIP64 npz/cnpy** binary incompatibility that had corrupted GPU-side hypervector encoding.
- Adapted the ANN-SoLo proteomics search tool to metabolomics and fixed a **LanceDB** query-explosion bug from non-unique GNPS identifiers, replacing a batch-by-identifier loop with an ordered sequential scan and cutting index construction from a failed state to seconds for ~1.5M spectra.
- Built a comparative pipeline benchmarking **HyperOMS** against **ANN-SoLo** on identical metabolomics queries, quantifying identification overlap to evaluate HDC-based search sensitivity versus cosine-based methods.

Co-Founder & CEO

Jan. 2026 – Present

Kyte - College Delivery Service

San Diego, CA

- Co-founded and lead a college delivery startup, owning business strategy, go-to-market, and product roadmap for a multi-sided marketplace (**customer, runner, admin** clients) built for UCSD's **40,000-student** campus.
- Engineered a concurrent backend with **Next.js**, **TypeScript**, **PostgreSQL (Neon)**, and **Redis/BullMQ**, building a fault-tolerant transactional engine with strict payment idempotency.
- Built **real-time state management** and **live dispatch routing** for meetpoint-based fulfillment, using **schema-driven API contracts** to target a **sub-25-minute** delivery SLA.

Software Engineering Intern

June 2025 – December 2025

Health1st.ai — WinFully on Technologies

Remote (Alpharetta, GA)

- Built **FastAPI** microservices handling **HL7**, **FHIR**, and **X12** healthcare data formats with async processing, **PostgreSQL**, and **Redis** caching, deployed to **GCP Cloud Run** via CI/CD.
- Developed an **AI mapping service** using **LangChain** and **GPT-4** to transform healthcare data between standard formats (HL7, FHIR, X12, CCDA).

PROJECTS

emnico 8-bit Accumulator Processor | *SystemVerilog, Quartus II, ModelSim, FPGA, Assembly*

Spring 2026

- Designed **emnico**, a custom **8-bit accumulator-style processor** with a fixed **9-bit instruction format**, optimized for three target CSE141L programs under a tight instruction-width budget.
- Implemented the CPU in **SystemVerilog**, including the program counter, instruction memory, control decoder, ALU, accumulator, address register, 8-register file, data memory, carry/zero flags, and done logic.
- Wrote a matching **assembler** to translate custom assembly into 9-bit machine code and verified program execution in **ModelSim** before synthesizing the design in **Quartus II** for an FPGA target.
- Hand-wrote assembly for Hamming distance, min/max absolute difference, and **signed 16×16 → 32-bit shift-and-add multiplication**, building 16-bit and 32-bit arithmetic from single-byte ALU operations using carry/borrow flags.

NBA Player Props EV Engine | *Python, NumPy, SciPy, Pandas*

Dec. 2025

- Engineered a **Monte Carlo simulation engine** (20K+ iterations) utilizing a **Student-t copula** ($\nu = 5$) to capture fat-tailed tail dependence, and applied Higham's algorithm for nearest-PSD correlation matrices.
- Modeled discrete prop probabilities via **Poisson** and **Negative Binomial distributions**, integrating a logit-space **Bayesian market blender** to anchor empirical projections with sharp sportsbook implied probabilities.

SKILLS AND COURSEWORK

Languages: Python, C, C++, Java, JavaScript, TypeScript, Swift, Bash, HTML/CSS

Hardware & Systems: SystemVerilog, Verilog, CUDA, FPGA (Intel Cyclone IV), Quartus II, ModelSim/Questa, Computer Architecture, Digital Design

Frameworks/Libraries: Next.js, FastAPI, React, Node.js, Prisma, Redis, NumPy, Pandas, SciPy, FAISS, LanceDB, Docker, PostgreSQL, MongoDB, AWS S3

Relevant Courses: Data Structures & OOP (CSE 12), Discrete Math (CSE 20), Systems Prog (CSE 29), Comp Architecture (CSE 30), Computer Architecture Lab (CSE 141L), Circuit Analysis (ECE 35 & 140), Digital Systems (CSE 140), Probability & Stats (ECE 109)